2013-2014 – Master 2 – Macro I

Lecture 6 : R&D based models of endogenous growth

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0. Introduction

- In neo-classical growth models technical progress is assumed to follow an exogenous trend. Therefore the model does not explain the long-run growth rate of the economy, which (if TP is labor-augmenting) must be equal to the growth rate of technology.
- The purpose of R&D - based models of growth is to endogenize technical progress. This is called endogenous growth theory.
- First generation endogenous growth models assumed that technical progress was the by product of economic activity by some mechanism such as learning by doing and productive externalities.
- In second generation models R&D is explicitly modeled as an activity which produces technical change and is driven by economic incentives.

1. On Knowledge

The state of technology is a form of knowledge. Knowledge has a number of specificities :

- It is non-rival : if I use a piece of knowledge, it does not prevent anybody else from using it. This is unlike a standard good like apples.
- Producers of apples own the apples ; they can make money from giving the apple to someone else in exchange for money.
- If the knowledge produced is visible, then because it is non rival the producer cannot in principle prevent others from using it even though they have not paid for it.
- That is, a standard good is naturally excludable. This is not the case for a nonrival good.
- In such a situation there are no private economic incentives for producing new knowledge and growth will stop.

1. On Knowledge

- However one can make knowledge artificially excludable by granting monopoly rights for its use, i.e. patents. (Other solutions are possible (public contests))
- This increases the incentives for innovation and thus boosts the growth rate.
- But it comes at the cost of monopoly price distortions.
- Trade-off between static and dynamic efficiency.
- For those reasons in endogenous growth models the equilibrium and the optimum differ : we cannot derive the equilibrium as an optimum, unlike in the Ramsey model.
- In the model we will study, the number of goods N_t is endogenous ; new goods will be introduced by an R&D sector, and this will be the source of growth.

2. A standard model of horizontal innovation

The consumer

- We consider a representative consumer whose utility is

$$\int_0^{+\infty} \frac{C_t^\alpha - 1}{\alpha} e^{-\rho t} dt.$$

- In that formula, C_t is a Dixit-Stiglitz aggregate of a continuum of individual goods of total mass N_t :

$$C_t = \left(\int_0^{N_t} c_{it}^\beta di \right)^{\frac{1}{\beta}},$$

where c_{it} is the individual's consumption of good i and $\beta \in (0, 1]$.

- From the lecture on Dixit-Stiglitz, aggregate price level :

$$p_t = \left(\int_0^{N_t} p_{it}^{1-\sigma} di \right)^{\frac{1}{1-\sigma}}, \qquad \text{(API)}$$

where

$$\sigma = \frac{1}{1-\beta}.$$

2. A standard model of horizontal innovation

The consumer (continued)

- We will take the aggregate consumption index as the numéraire. Therefore at any point in time

$$p_t = 1.$$

- We know that the value of total consumption expenditures is $p_t C_t$. Hence it is equal to C_t.
- One can borrow and lend at an instantaneous rate r_t. That is, there exist instantaneous bonds that pay $1 + r_t dt$ units of the numéraire at $t + dt$ in exchange for one unit of the numéraire at t.

2. A standard model of horizontal innovation

The consumer (continued)

- Borrowers issue those bonds and lenders buy them.
- The instantaneous interest rate r_t adjusts so that there is equilibrium on the market for loanable funds, i.e. the supply of bonds is equal to the demand for bonds.
- The consumer has a flow of income denoted by Y_t. He is endowed with L units of labor (equivalently L can be interpreted as the size of the population), and he owns the firms in the economy.
- Denoting by w_t the wage at date t, we have that

$$Y_t = w_t L + \Pi_t,$$

  where Π_t are aggregate profits.
- Let W_t be the consumer's net wealth at t : this is the total value of the bonds held by the consumer. It evolves according to (remember that $p_t = 1$).

$$\dot{W}_t = r_t W_t + Y_t - C_t \qquad \text{(IBC)}$$

2. A standard model of horizontal innovation

The consumer (continued)

- The consumer problem can be decomposed in two stages :
  1. For a given C_t, what is my optimal allocation of C_t across goods ? Answer from the Dixit-Stiglitz block.
  2. What is the optimal intertemporal allocation of C_t ?
- To answer question 2, we integrate backward the instantaneous budget constraint (IBC) and get

$$W_t = W_0 e^{\int_0^t r_u du} + \int_0^t (Y_u - C_u) e^{\int_u^t r_v dv} du, \qquad (1)$$

  where W_0 is the initial wealth of the consumer, inherited from the past.
- Exercise 1 : Derive this by noting that (IBC) is a linear differential equation and that the above is the unique solution such that $W_{t=0} = W_0$. [Hint : Derive this by making use of the following quantity : $\tilde{W}_t = W_t e^{-\int_0^t r_u du}$ and by re-expressing the IBC in terms of $\frac{d}{dt}\tilde{W}_t$.]

2. A standard model of horizontal innovation

The consumer (continued)

- We impose the solvency condition

$$\lim_{t \to +\infty} W_t e^{-\int_0^t r_u du} \geq 0.$$

- Substituting into (1), this delivers the intertemporal budget constraint of the consumer

$$W_0 + \int_0^{+\infty} (Y_t - C_t) e^{-\int_0^t r_v dv} dt \geq 0. \qquad \text{(ITBC)}$$

- The present discounted value of net borrowing $C_t - Y_t$ cannot exceed initial wealth.

2. A standard model of horizontal innovation

The consumer (continued)

- We are now in a position to write the Lagrangian

$$\mathcal{L} = \int_0^{+\infty} \frac{C_t^\alpha - 1}{\alpha} e^{-\rho t} dt + \lambda \left(W_0 + \int_0^{+\infty} (Y_t - C_t) e^{-\int_0^t r_v dv} dt \right).$$

- The FOC is

$$C_t^{\alpha-1} e^{-\rho t} = \lambda e^{-\int_0^t r_v dv}.$$

- Taking log derivatives with respect to time we eliminate λ and get the Euler equation

$$\frac{\dot{C}_t}{C_t} = \eta(r_t - \rho), \tag{EE}$$

where

$$\eta = \frac{1}{1-\alpha}.$$

2. A standard model of horizontal innovation

The consumer (continued)

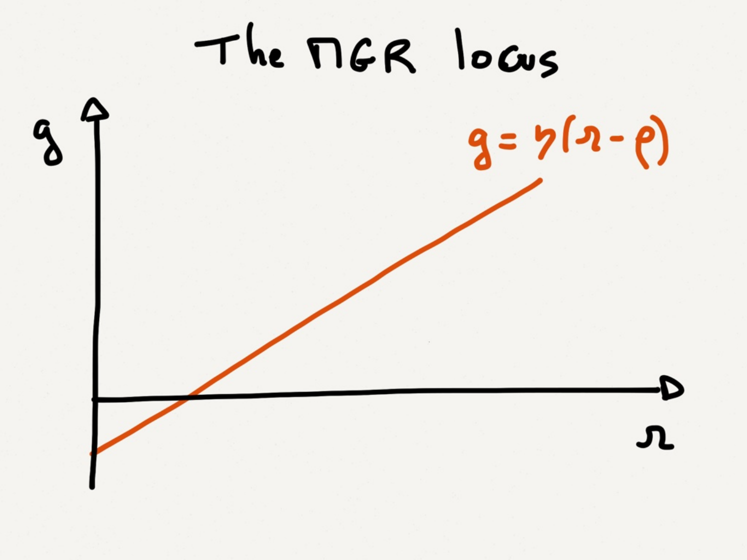
- Consider a BGP where C grows at rate g. Then along that BGP r must be constant and (EE) implies a positive relationship between r and g :

$$g = \eta(r - \rho). \tag{MGR}$$

- The greater the growth rate, the greater must the interest rate be to induce consumers to plan for a greater consumption in the future relative to now.
- The response of r to g is smaller, the greater η : The more I am willing to substitute intertemporally, the smaller the increase in r that will induce me to postpone consumption by a given relative amount.
- This is just the modified Golden rule but g is now treated as endogenous.

2. A standard model of horizontal innovation

The consumer (continued)



2. A standard model of horizontal innovation

The production sector

- Individual goods are produced using one unit of labor, i.e.

$$y_i = l_i.$$

- Consequently, the cost function of firm i at t is

$$c_t(y_i) = w_t y_i.$$

- From the dixit stigliz block we then get that

$$p_{it} = \mu w_t = p_{0t}, \quad \text{with} \quad \mu = \frac{\sigma}{\sigma - 1}.$$

- Using (API) we have that

$$p = 1 = p_{0t} N_t^{\frac{1}{1-\sigma}} = \mu w_t N_t^{\frac{1}{1-\sigma}}.$$

- That is

$$w_t = \frac{N_t^{\frac{1}{\sigma-1}}}{\mu}. \tag{RW}$$

2. A standard model of horizontal innovation

The production sector (continued)

Interpretation :

- w_t is the consumption wage.
- When there are more products, the consumption wage goes up. This is because one unit of labor produces a greater variety of goods, and therefore more units of welfare.
- The effect is stronger, the greater the taste for variety, i.e. the smaller σ.
- If goods are perfect substitutes ($\sigma \to \infty$), new goods do not bring any extra welfare and the consumption wage does not go up with N
- If μ is larger, goods are more expensive relative to the wage and the equilibrium wage falls.

2. A standard model of horizontal innovation

The innovation sector

- New goods are invented using labor.
- At any point in time a fraction of the workforce is employed in the R&D sector.
- To invent 1 new good per unit of time I need $1/(\gamma N_t)$ units of labor.
- Here N_t stands for an externality : for N to grow in a sustained way, we need the cost of producing new goods to fall over time. Otherwise we will just be able to produce a constant flow of new goods, which will fall relative to N, implying that $\dot{N}/N$ will gradually fall to zero over time.
- The flow of new goods is

$$\dot{N}_t = \gamma N_t L_{Rt},$$

where L_{Rt} is total employment in the R&D sector.

2. A standard model of horizontal innovation

The innovation sector (continued)

- How much R&D do firms do ? For this we have to compute the costs and benefits of R&D.
  - The costs are given by the wage that must be spent for inventing one extra good. This is equal to
  $$\frac{w_t}{\gamma N_t}.$$
  - The benefit is given by the present discounted value of the profits from innovation. We assume an innovator gets a patent entitling it to the whole stream of monopoly profits associated with the new good. Let V_t be the value of this patent. Then
  $$r_t V_t = \pi_t + \dot{V}_t, \tag{PV}$$

  where π_t is a firm's profit at date t, given by
  $$\pi_t = (\mu - 1) w_t y_t. \tag{PR}$$
- Here x_t is the output of any individual firm (by symmetry these are the same across firms).

2. A standard model of horizontal innovation

The innovation sector (continued)

- In equilibrium, entry into the R&D sector occurs until the cost of an innovation equals its benefit.
- Therefore we must have

$$V_t = \frac{w_t}{\gamma N_t}. \tag{2}$$

- Exercise 2 : One may now make use of the condition from Dixit-Stiglitz which says that $y_t = C_t(\mu w_t)^{-\sigma}$. Show that using this condition delivers (RW) again.

2. A standard model of horizontal innovation

Equilibrium

- In equilibrium, we must have that

$$L = y_t N_t + L_{Rt}.$$

- Consequently,

$$y_t = \frac{L - L_{Rt}}{N_t} = \frac{L - \frac{1}{\gamma}\frac{\dot{N}_t}{N_t}}{N_t}. \tag{3}$$

- Substituting this and (RW) into (PR) we get

$$\pi_t = (\mu - 1)\frac{N_t^{\frac{2-\sigma}{\sigma-1}}}{\mu}\left(L - \frac{1}{\gamma}\frac{\dot{N}_t}{N_t}\right). \tag{4}$$

2. A standard model of horizontal innovation

Equilibrium (continued)

- Consider now a BGP. In a BGP, the numerator of the RHS of (3) is constant. Therefore,

$$g_y = -g_N.$$

- Total output in physical terms is Ny and is constant over time. This is normal since there is no physical productivity growth. But total welfare grows over time because of the taste for variety.
- Furthermore, $C_t = y_t N_t^{\frac{1}{\beta}} = y_t N_t^{\frac{\sigma}{\sigma-1}}$.
- Therefore,

$$g_C = g = g_y + \frac{\sigma}{\sigma - 1} g_N = \frac{1}{\sigma - 1} g_N.$$

- Hence the growth of welfare is larger in proportion to g_N, the lower σ,i.e. the more the new goods are complementary with the existing ones and therefore "useful".

2. A standard model of horizontal innovation

Equilibrium (continued)

- Hence $g_N = (\sigma - 1)g$.
- Remark 1 : Note that along a BGP, $g_w = \frac{g_N}{\sigma-1} = g_{wL} = g_{wL+\Pi} = g$. Thus wages and nominal income also grow at g. And since $p = 1$ throughout, this is also true of real income.
- This helps us to rewrite (4) :

$$\pi_t = (\mu - 1)\frac{N_t^{\frac{2-\sigma}{\sigma-1}}}{\mu}\left(L - \frac{\sigma - 1}{\gamma}g\right).$$

- Note that $g_\pi = g_V = \frac{2-\sigma}{\sigma-1}g_N = (2-\sigma)g$.
- If $1 < \sigma < 2$, profits grow over time : the quantity of a given good being produced falls less fast than its price. The more goods are substitute, the lower the growth rate of their price (and of wages), and the less profits grow over time. For $\sigma > 2$ profits fall over time.

2. A standard model of horizontal innovation

Equilibrium (continued)

- In a BGP, (PV) implies that

$$V_t = \frac{\pi_t}{r - g_V} = \frac{(\mu - 1)}{r - (2 - \sigma)g} \frac{N_t^{\frac{2-\sigma}{\sigma-1}}}{\mu} \left(L - \frac{\sigma - 1}{\gamma} g \right).$$

- In equilibrium this must be equal to

$$\frac{w_t}{\gamma N_t} = \frac{1}{\gamma N_t} \frac{N_t^{\frac{1}{\sigma-1}}}{\mu} = \frac{N_t^{\frac{2-\sigma}{\sigma-1}}}{\gamma \mu}.$$

- Equating the two expressions delivers another equilibrium relationship between r and g

$$\frac{(\mu - 1)}{r - (2 - \sigma)g} \left(L - \frac{\sigma - 1}{\gamma} g \right) = \frac{1}{\gamma},$$

- or equivalently (given that $\mu = \frac{\sigma}{\sigma - 1}$)

$$g = \frac{(\mu - 1)\gamma L - r}{\sigma - 1}. \tag{RD}$$

2. A standard model of horizontal innovation

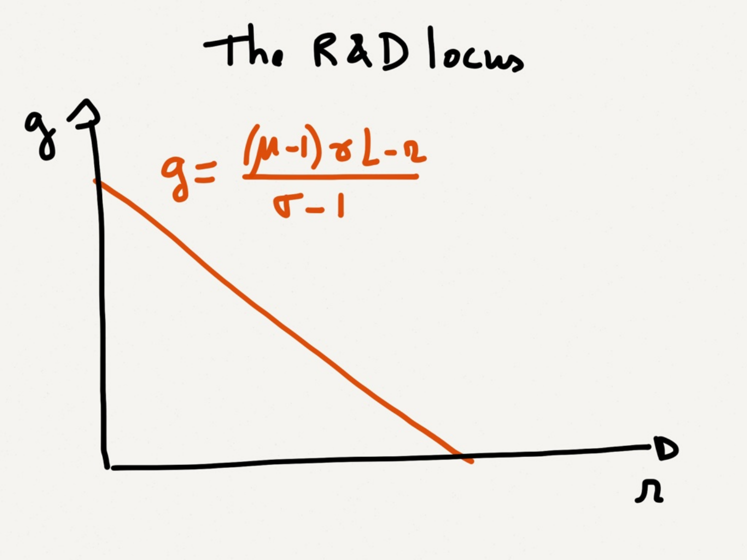
Equilibrium (continued)

$$g = \frac{(\mu - 1)\gamma L - r}{\sigma - 1} \qquad (RD)$$

- This now defines a negative relationship between g and r. We have made use of many equilibrium conditions to derive it. But it is chiefly driven by the incentives to do R&D.
- g falls with r : a greater cost of capital discourages innovation and thus growth.
- g grows with μ (falls with σ). The mechanism is two-fold. Greater markups raise profitability, thus increasing incentives to innovate. Greater complementarities make g greater given g_N.
- g grows with γ : less costly R&D increases innovation.
- g grows with L : more people around increase the equilibrium number of innovators and therefore the flow of new ideas.

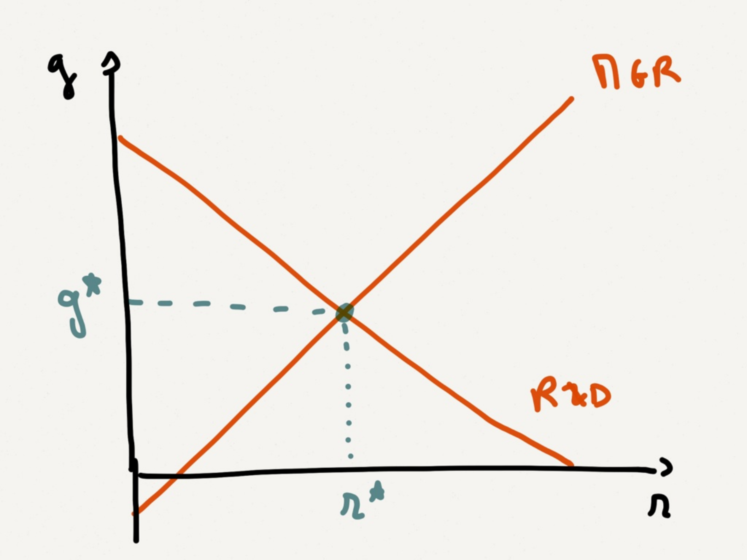
2. A standard model of horizontal innovation

Equilibrium (continued)



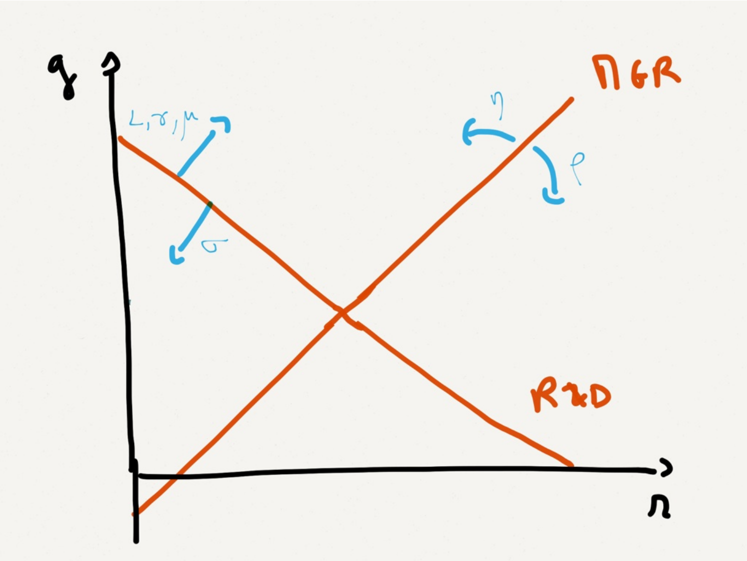
2. A standard model of horizontal innovation

Equilibrium (continued)



2. A standard model of horizontal innovation

Equilibrium (continued)



2. A standard model of horizontal innovation

Equilibrium (continued)

- Equilibrium in a BGP is determined by the intersection between (RD) and (MGR)
- This delivers a growth rate g and and interest rate r.
- The comparative statics and their interpretation are straightforward.
- Algebraically, we get

$$
\begin{aligned}
g &= \frac{\eta}{(\sigma-1)\eta+1}\left[(\mu-1)\gamma L-\rho\right];\\
r &= \frac{(\mu-1)\gamma L+(\sigma-1)\eta\rho}{(\sigma-1)\eta+1}.
\end{aligned}
$$

3. Some Remarks

Remark 1 : The equilibrium is not efficient

There is a number of reasons for that :

- Reason 1 : The monopoly distortion in price-setting tends to push prices above marginal cost. In equilibrium this means that real wages are too low relative to the social optimum.
- Reason 2 : Innovators do not entirely appropriate the social surplus from their innovation. They get profits but part of the surplus goes to the consumers. This tends to make innovation too low compared with the optimum.

3. Some Remarks

Remark 1 : The equilibrium is not efficient (continued)

Note :

- Reason 1 goes in the opposite direction of reason 2 : lower wages reduce the cost of inventing new goods, so it boosts R&D.
- Another way to look at it is to say that the only alternative to work in the productive sector is the R&D sector.
- Therefore, any distortion that reduces incentives to work in the production sector (such as the monopoly markup here) tends to be good for R&D.
- Things would be different if for example labor supply were elastic. In this case the monopoly markup would reduce total labor supply.
- Remark : In some models the two effects exactly cancel out and the equilibrium is optimal.

3. Some Remarks

Remark 1 : The equilibrium is not efficient (continued)

- Reason 3 : There is a productive externality in the production of new goods. Innovators do not internalize the gains from the fact that their invention increases the productivity in inventing further goods.
- Remark : While such an externality is plausible, remember we introduced it essentially for technical reasons, i.e. to get sustained growth.

3. Some Remarks

Remark 2 : scale effect

- The model exhibits *scale effects :* larger economies (i.e. with a greater L) grow faster, as the above formulae make clear.
- This is because the flow of innovation depends on the number of people doing R&D, which is itself proportional to population size. Thus larger populations lead to more innovation and faster growth of GDP per capita.
- There is much debate on whether that is realistic.